

HW07

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Exercise 2

- Points attempted: 5
 - Self-assessed score: 5/5.
 - Time spent: 60 minutes.
 - Difficulty: 6/10.
 - Questions/Feedback: Clarity, correctness.
1. [2] Read Tai-Danae Bradley's blog post What's a Quotient Group, Really? Part 1. (Professor Bradley wrote this blog when she was in graduate school, and her blog was a part of the motivation for doing a blog post as the term paper in this class.)

Done.

2. HW7, Exercise 2: i added the defintion of an abelian group.
3. Do at least *three* of the following for your blog:

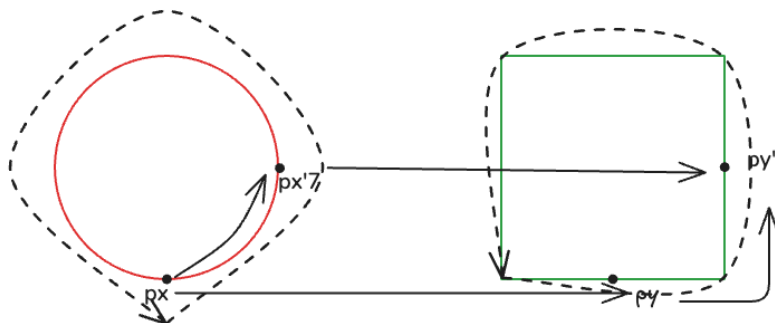
- a Write an outline of what your blog post is going to be about.

Although the full prompt that I will be following is "The Fundamental Group of a Topological Space", I want to maybe open up the scope just a tad, and include more details about algebraic topology itself. Thus, I want to include a section on specifically the foundations of algebraic topology. Then, as a we build solid foundation, the blog should transition into a more focused discussion about how groups show up in algebraic topology, specifically as the fundamental group of a topological space.

Lastly, I want to also get into some applications. As I am currently deeply interested in machine learning, and the mathematical foundations of it, I want to explore some applications of topology in machine learning. There is also a concept in deep learning known as the **manifold hypothesis** which states that 'most' high dimensional data can actually be summarized as a manifold with a lower 'intrinsic dimensionality.'

- b Write two paragraphs explaining a core concept or idea.

The bedrock of topology is a concept known as **homotopy**. Although topology is often introduced in terms of donuts and mugs, I believe that a slightly more 'formal' defintion is worth investing into before we continue. Without too much detail[TK], suppose we have two topological spaces X and Y . Now, consider two maps from $f, f': X \rightarrow Y$. We consider f and f' **homotopic** if we can "continuously deform" one into the other. This may be hard to visualize, but imagine that we have two topological spaces, one that is the perimeter of a circle, and another that is the perimeter of a square, (we will see later that these two are inherently the same space, but for now, we can treat them as different spaces). Notice that all mappings from a circle to a square can be deformed into each other, if they have a specific quality: that they go around the object the same amount of times.



Imagine a mapping where we go around the circle once, but around the square twice. [TK] No matter how we stretch this mapping, we can never get something that maps from the circle to the square that go around the same number of times. The formal name for this concept is a **winding number**. Thus, all mappings in this setting are homotopic if they share a winding number. Further more, we can group the mappings that are homotopic [TK, need to explain why homotopy is an equivalence relation] into something known as **homotopy classes**.

- c Include an illustration, or plan how you will make your illustration.

A made a rough illustration of the example above.

Exercise 3

- Points attempted: 6
- Self-assessed score: 6/6.
- Time spent: 60 minutes.
- Difficulty: 4/10.
- Questions/Feedback: Clarity, correctness.

Let G be a group. Then we say that $a \in G$ is conjugate to $b \in G$ if there exists $g \in G$ such that $a = gbg^{-1}$. The relation is conjugate to is an equivalence relation, so it allows us to partition G into equivalence classes (which we call conjugacy classes). Intuitively, a and b are conjugate if performing a from one perspective is the same as performing b from another perspective. As an example, a horizontal flip and a vertical flip of a square are conjugate in D_4 because if you're standing 90° to the right of me and I do a horizontal flip of the square, it will look like a vertical flip from your perspective—but a horizontal flip to me will never look like a 180° rotation to you. In the following exercises you will be looking at conjugacy classes of the symmetry group of the hexagon.

1. Prove that e is in its own conjugacy class.

Proof. For all $g \in G$, $geg^{-1} = gg^{-1} = e$. Therefore, e is only ever conjugate to itself, thus it is its own conjugacy class. $\square$

2. Find an example of $g \in D_6$ such that $r^2 = gr^4g^{-1}$.

Let $g = f$. Now, we have the equation $r^2 = fr^4f^{-1} = fr^4f = ffr^2 = r^2$.

3. Find an example of $g \in D_6$ such that $f = gr^2fg^{-1}$.

First, we feel that g is of the form r^k . Thus, we can write the equation $f = r^k r^2 f r^{-k}$. We can use some algebra to manipulate the equation to solve for k , and thus g .

$$f = r^k r^2 f r^{-k} \quad (1)$$

$$f = r^{k+2} f r^k \quad (2)$$

$$f = r^{2k+2} f \quad (3)$$

$$r^{2k+2} = e \quad (4)$$

$$2k + 2 = 0, 6 \quad (5)$$

$$k = -1, 2 \quad (6)$$

We see that for both $g = r^{-1} = r^5$ and $g = r^2$, $f = g r^2 f g^{-1}$.

4. Prove that if $\phi: D_6 \rightarrow H$ is a homomorphism, and g_1 is conjugate to g_2 in D_6 , then $\phi(g_1)$ is conjugate to $\phi(g_2)$ in H .

Proof. First, we see that because g_1 is conjugate to g_2 , there exists some $d \in D_6$ such that $g_1 = d g_2 d^{-1}$. Since ϕ is a homomorphism, it must preserve structure.

Therefore,

$$\phi(g_1) = \phi(d g_2 d^{-1}) \quad (7)$$

$$= \phi(d) \phi(g_2) \phi(d^{-1}) \quad (8)$$

$$= \phi(d) \phi(g_2) \phi(d)^{-1} \quad (9)$$

$$(10)$$

Since $\phi(d) \in H$, $\phi(g_1)$ must be conjugate to $\phi(g_2)$. $\square$

5. Prove that if $\phi: D_6 \rightarrow H$ is a homomorphism, H is an abelian group, and g_1 is conjugate to g_2 in D_6 , then $\phi(g_1) = \phi(g_2)$.

Proof. Following from the above proof, we have $\phi(g_1) = h \phi(g_2) h^{-1}$ for some $h \in H$. Now, because H is abelian, we can rearrange the RHS to be $h h^{-1} \phi(g_2) = e_H \phi(g_2)$. Therefore, $\phi(g_1) = \phi(g_2)$. $\square$

6. Let $\Phi: D_6 \rightarrow \mathbb{Z}/2\mathbb{Z}$ be defined by $\Phi(r) = 0$ and $\Phi(f) = 1$, and use this to show that $r^n f$ is not conjugate to r^m for any integers n and m .

First, we show that Φ is a homomorphism by looking at the group presentations of D_6 and the $\text{Im}_\Phi(D_6)$.

$$D_6 = \langle r, f \mid r^6 = f^2 = (rf)^2 = e \rangle$$

$$\text{Im}_\Phi(D_6) = \langle 0, 1 \mid 0^6 = 1^2 = (01)^2 = e \rangle$$

These group presentations show that Φ is structure preserving, therefore Φ is a homomorphism.

Now, if Φ is a homomorphism, then conjugacy is preserved across the mapping. In other words, if $r^n f = g r^m g^{-1}$, then $\Phi(r^n f) = \Phi(g) \Phi(r^m) \Phi(g)^{-1}$.

Now, we use the group presentation of $\text{Im}_\Phi(D_6)$ to simplify to

$$1 = h 0 h^{-1}.$$

Because $\mathbb{Z}/2\mathbb{Z}$ is abelian, we have a contradiction, therefore $r^n f$ can not be conjugate to r^m .

7. Let $\psi: D_6 \rightarrow \mathbb{Z}/2\mathbb{Z}$ be defined by $\psi(r) = 1$ and $\psi(f) = 0$, and use this to prove that none of the elements in $\{r, rf, r^3, r^3f, r^5, r^5f\}$ are conjugate to any of the elements in $\{e, f, r^2, r^2f, r^4, r^4f\}$.

Similar to the previous proof, we examine the group presentation of the image of Φ . Notice that

$$\text{Im}_\Phi(D_6) = \langle 0, 1 \mid 1^6 = 0^2 = (01)^2 = e \rangle$$

shows that Φ is a homomorphism.

Now, let $A = \{r, rf, r^3, r^3f, r^5, r^5f\}$, and $B = \{e, f, r^2, r^2f, r^4, r^4f\}$. Now, using the group presentation, notice that $\Phi(a) = 1$ for all $a \in A$, and $\Phi(b) = 0$ for all $b \in B$. 0 and 1 are not conjugate in $\mathbb{Z}/2\mathbb{Z}$, therefore none of the elements of A are conjugate to any of the elements of B .

8. Let $\varphi: D_6 \rightarrow S_3$ be the homomorphism defined by $\varphi(f) = (1\ 2)$ and $\varphi(r) = (1\ 2\ 3)$. Note that S_3 is not an abelian group, but use this homomorphism to show that $r^3 \in D_6$ is in its own equivalence class.

First, we see that $\varphi(r^3) = e_{S_3}$. Since $r^3 \in \ker(\varphi)$, r^3 must be in its own conjugacy class.

9. Partition the dihedral group of the hexagon

$$D_6 = \langle r, f \mid r^6 = f^2 = (rf)^2 = e \rangle$$

into conjugacy classes. You don't have to prove every piece of this, but you do have to make a note of any claims that you made that you didn't prove.

First, we showed that $\{e\}$ is its own conjugacy class. Similarly, $\{r^3\}$ is its own conjugacy class. (I don't think we actually proved that the elements of a kernel are in their own conjugacy class).

Then, we see that A and B can potentially be conjugacy classes, but we also showed that elements of the form $r^n f$ can not be conjugate to elements of the form r^m .

Therefore, we can partition A and B into the sets: $\{r^1, r^5\}, \{r^2, r^4\}, \{f, r^2f, r^4f\}, \{rf, r^3f, r^5f\}$.

Choice Exercise 5

- Points attempted: 13
 - Self-assessed score: 11/13.
 - Time spent: 100 minutes.
 - Difficulty: 4/10.
 - Questions/Feedback: Clarity, correctness.
1. [3] Attempt the following true/false questions and give brief explanations for your reasoning.
- (a) If $\varphi: D_8 \rightarrow D_4$ is a homomorphism, then

$$D_8 / \ker(\varphi) \cong D_4.$$

True. φ is structure preserving, so the quotient group between D_8 and the kernel of φ must be of the same structure as D_4 .

- (b) If G is an infinite group with $H \leq G$, then the index of H in G is finite: $[G : H] < \infty$.

False. The group $H = C_n$ for $n \in \mathbb{N}$ is a subgroup of the "continuous" cyclic group $G = \mathbb{R}/\mathbb{I}$, however the index $[G : H]$ is uncountably infinite.

- (c) Let $G \times H$ be a direct product of groups, and let

$$N = \{(g, e_H) \mid g \in G\} \cong G.$$

Then $N \triangleleft G \times H$ and the quotient group $(G \times H)/N$ is isomorphic to H .

True. N is a normal subgroup because the ‘left’ component is just the entire group G . A group is always a normal subgroup of itself. Then the ‘right’ component is the trivial group, which is a normal subgroup of any group, so it is a normal subgroup of H . Therefore, $N \triangleleft G \times H$. The quotient group can then be “distributed” to result in a group where the ‘left’ component is the trivial group, and the ‘right’ component is just H . This group is isomorphic to H .

- (d) If G is a group, and N is a normal subgroup, then there exists a subgroup H in G such that $G/N \cong H$.

True. We can pick a ‘representative’ from each of the cosets of N in G to form H .

- (e) If $\varphi: G \rightarrow H$ is an homomorphism with kernel $K = \ker(\varphi)$, then the map $\psi: G/K \rightarrow \text{Im}(\varphi)$ that sends $\psi(aK) = \varphi(a)$ is also a homomorphism.

True. ψ honestly just feels equivalent to φ .

- (f) If $g \in G$ has order k and $h \in H$ has order k , then there exists an isomorphism $\varphi: \langle g \rangle \rightarrow \langle h \rangle$.

True. Both $\langle g \rangle$ and $\langle h \rangle$ are isomorphic to C_k .

- (g) If the only normal subgroups of G are $\{e\} \triangleleft G$ and $G \triangleleft G$, then G is the trivial group (i.e. $|G| = 1$.)

False. It feels like there *could* exist some counter example of a group that has no non-trivial normal subgroups.

- (h) If G is an abelian group and $H \leq G$, then $H \triangleleft G$ and G/H is abelian.

True. If the elements of G commute, then the elements of H must also commute. Similarly, by definition of normal sub groups, the cosets are just elements from the group G . Therefore, they are also commutative/abelian.

- (i) If G and H are isomorphic, then any homomorphism from G to H is an isomorphism.

False. We can always construct the ‘trivial homomorphism’ that sends all elements of G to e_H .

- (j) If G is a finite group with $|G| \geq 3$ and H is an infinite group, then any homomorphism $\varphi: G \rightarrow H$ fails to be one-to-one.

False. This is from set theory.

2. Expand the section below to see the answers. If you got the answer or explanation wrong, write the correct answer here and say something about why you got it wrong originally.

I got two wrong: (a) and (d). For (a), I just failed to consider the case of the trivial group. However, for (d), my line of reasoning was fundamentally flawed. I am still not too sure of my reasoning, so I will have to go back through the warmup mentioned in the solution.

Choice Exercise 7

- Points attempted: 5
- Self-assessed score: 5/5.
- Time spent: 30 minutes.
- Difficulty: 6/10.
- Questions/Feedback: Clarity, correctness.

2. [5] In this part, you’ll study some subgroups and quotient groups of this group, which might not be as hard as you expect!

- (a) Write some elements of the cyclic subgroup $H = \langle ab \rangle \leq F_2$.
 We have elements like e , ab , $(ab)^{-1} = b^{-1}a^{-1}$, $abab$, and $abab \dots ab$.
- (b) Is $H \triangleleft F_2$? Why or why not?
 H can not be a normal subgroup of F_2 because for $b \in F_2$, $bH \neq Hb$. We can see this because H can not include an element that starts with b , however all elements of bH start with b .
- (c) Consider the map $\varphi: F_2 \rightarrow D_3$ by $\varphi(a) = r$ and $\varphi(b) = f$. Is this map a homomorphism? An isomorphism?
 First, this map can't be an isomorphism because $|F_2| \neq |D_3|$. At most, this map can be a homomorphism. We also see that φ is actually a homomorphism because for any $x, y \in F_2$, $\varphi(xy) = \varphi(x)\varphi(y)$.
- (d) What is the image of φ ?
 The image of φ is the entire D_3 .
- (e) Without explicitly describing $\ker(\varphi)$, what does the first isomorphism theorem have to say about the quotient group $F_2/\ker(\varphi)$?
 The first isomorphism theorem says that there exists an isomorphism between $\psi: F_2/\ker(\varphi) \rightarrow D_3$.